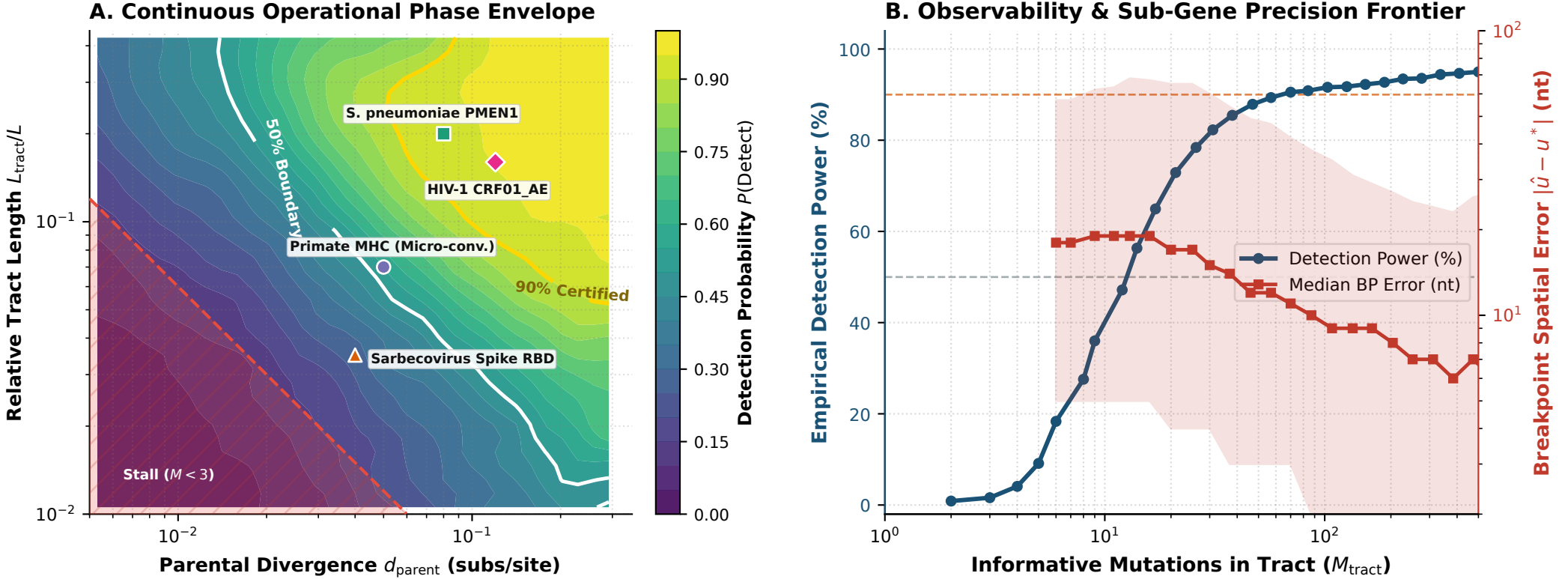
A. Continuous Operational Phase Envelope
S. pneumoniae PMEN1
HIV-1 CRF01_AE
Primate MHC (Micro-conv.)
Sarbecovirus Spike RBD
50% Boundary
90% Certified
Stall ($M < 3$)
Relative Tract Length $L_{\text{tract}}/L$
Parental Divergence $d_{\text{parent}}$ (subs/site)
Detection Probability $P$(Detect)
B. Observability & Sub-Gene Precision Frontier
Empirical Detection Power (%)
Breakpoint Spatial Error $|\hat{u} - u^*|$ (nt)
Informative Mutations in Tract ($M_{\text{tract}}$)
Detection Power (%)
Median BP Error (nt)